

Bayesian end-of-day rebalancing of a bike-sharing system

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1 Introduction

Bike sharing is a cheap and flexible way to move around a city, but it only works when there are bikes where people want them. Trips usually go one way: in the morning people ride from home areas into the centre, and in the evening they ride back. Because of this, some stations keep running empty while others fill up. Operators fix this by *rebalancing*, that is, moving bikes between stations (mostly at night) so that each station starts the next day with a sensible number of bikes. Rebalancing is the biggest running cost of such a system. So it helps a lot to know which stations will gain or lose bikes over a day, how many, and how sure we can be about it.

2 Research question and data

Our question is: *for each station, what does the daily net flow (returns minus rentals) look like, and how many bikes have to be moved at the end of the day?* We use the open **Citi Bike** trip data for New York City (Citi Bike 2026; citibikenyc.com/system-data, raw files at s3.amazonaws.com/tripdata). Each trip record gives the start and end station (with IDs and coordinates), the start and end times, and whether the rider is a member or a casual user. For every station s and day t we count the departures $D_{s,t}$ (trips that start at s) and the arrivals $A_{s,t}$ (trips that end at s). The net flow is $N_{s,t} = A_{s,t} - D_{s,t}$.

3 Past work and shortcomings

Most studies, including an earlier project in this seminar, predict the *total* number of rentals from the weather, using machine learning or a Bayesian count model (for example a Negative-Binomial regression of daily rentals on temperature and rain). These tell you how many trips happen in total, but they ignore where trips start and end. So they cannot say where bikes pile up or run out, which is exactly what rebalancing needs. There is also a lot of work on rebalancing itself, but it is usually written as an optimisation problem that takes a fixed demand estimate as input (Chemla et al. 2013; O'Mahony & Shmoys 2015). It works with single-number forecasts and gives no measure of uncertainty. Neither approach gives a full distribution of the signed imbalance at each station.

4 Method

We treat departures and arrivals as events that happen over a fixed time, so we model both with **Poisson** regressions and a log link,

$$D_{s,t} \sim \text{Poisson}(\lambda_{s,t}^{\text{dep}}), \quad A_{s,t} \sim \text{Poisson}(\lambda_{s,t}^{\text{arr}}), \quad \log \lambda_{s,t}^{\bullet} = \alpha_s^{\bullet} + \sum_j \theta_j^{\bullet} x_{tj}.$$

Each station gets its own intercept $\alpha_s^{\bullet}$ (with a shared, hierarchical Normal prior, so stations borrow strength from each other), and x holds calendar variables (weekday or weekend, time-of-day block). The log link keeps the rates positive. The net flow is the difference of two Poisson counts, which follows a **Skellam distribution**, $N_{s,t} \sim \text{Skellam}(\lambda_{s,t}^{\text{arr}}, \lambda_{s,t}^{\text{dep}})$, with mean $\lambda^{\text{arr}} - \lambda^{\text{dep}}$ and variance $\lambda^{\text{arr}} + \lambda^{\text{dep}}$ (Skellam 1946). Skellam is the right choice here because the net flow is a whole number that can be positive (a station filling up) or negative (a station emptying). A plain Poisson or Negative-Binomial can never be negative, but Skellam can (Fig. B). We fit the model in **PyMC** with the NUTS sampler (Salvatier et al. 2016) and reuse the Poisson and regression tools from the seminar (Downey 2021); the net-flow prediction is then read off the two fitted Poisson parts. What is new compared to earlier work is that we keep the station-level detail and give, for each station, a full predictive distribution of the overnight imbalance instead of one demand number.

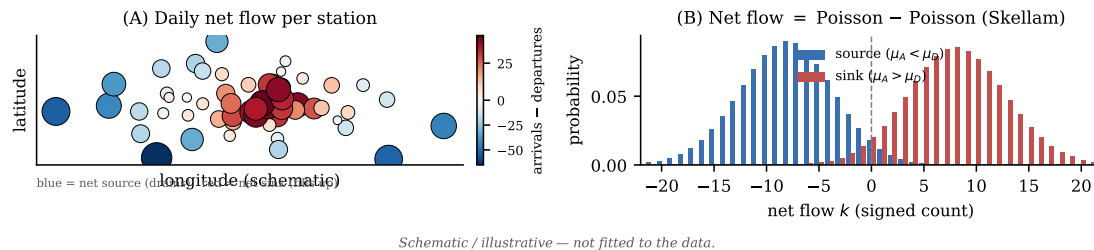


Figure 1: Sketch of the idea. (A) stations coloured by their daily net flow show a source/sink pattern; (B) the net flow as a Skellam (Poisson minus Poisson) distribution, for one emptying and one filling station.

5 Expected outcome / working hypothesis

We expect a clear pattern on the map: stations in home areas on the edge tend to lose bikes (net sources), while central and transit stations tend to gain them (net sinks), and the direction flips between morning and evening. We also expect most of the imbalance to come from a small number of stations. The Skellam prediction should give each station an overnight imbalance with a credible interval, wider for busy stations (the variance is the sum of the two rates). This lets us say things like $P(N_s < -c)$ for some safety buffer c . We will check convergence with the usual MCMC diagnostics ($\hat{R} \approx 1$, enough effective samples, clean trace plots). We do not have to solve the problem yet.

References

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